INDEX

Sr. No.	Experiment	Date	Signature

Job 1

Aim: Disassembly, Component Identification, and Reassembly of a Desktop Computer

Theory:

What is a computer system?

A computer system is a programmable electronic device that can accept input, store data, process that data according to instructions, and produce output. It consists of two main parts: hardware and software.

- Hardware components are the physical parts of the system, including the central processing unit (CPU), memory (RAM and storage devices like hard drives or SSDs), input devices (like keyboard and mouse), output devices (like monitor and printer), and other peripherals.
- Software components are the programs and operating systems that run on the hardware. The software directs the hardware on how to perform tasks. It includes system software (such as the operating system) that manages the hardware and application software that allows users to perform specific tasks (e.g., word processing, browsing).

Tools Required to Disassemble a Desktop Computer

- Phillips-head screwdriver (essential for most screws in a computer case)
- Flat-head screwdriver (sometimes required for prying open or removing certain components)
- Gloves (optional, protect hands from sharp edges)
- Anti-static wrist strap (to prevent electrostatic discharge from damaging components)
- Smartphone or notepad (to document steps for easier reassembly)
- Plastic cup or box (to safely keep screws, nuts, bolts)
- Compressed air or soft brush (for cleaning dust during disassembly)

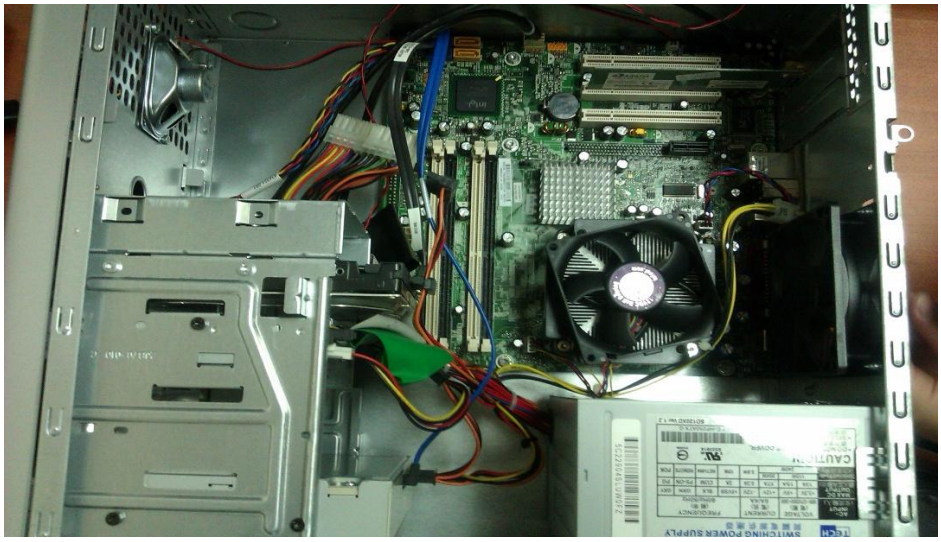


Figure 1 Desktop

Disassembly of a Desktop computer

1) Power Down and Unplug
    - Shut down the system completely.
    - Unplug all cables: power cord, monitor, keyboard, mouse, etc.
2) Discharge Static electricity
    - Wear an anti-static strap or touch an unpainted metal part of the case to prevent static damage.
3) Open the case
    - Place the cabinet on a flat and stable surface.
    - Unscrew and remove the side panels using a screwdriver.
    - Keep screws safe in a box.
4) Disconnect all internal cables
    - Unplug all power and data cables from the motherboard, graphics card, storage devices, and other components.
5) Remove Case Fans
    - Unplug and unscrew the case fans from their mounts, and gently lift them out.
6) Remove Expansion cards
    - Release any clips and unscrew expansion cards like graphics or sound cards.
    - Gently pull out cards from the motherboard slots.

7) Remove Storage Devices
    - Unscrew and carefully remove the drives from their bays.
8) Remove Ram
    - Push the clips at both ends of the RAM slots, and gently lift out RAM sticks without touching the connectors.
9) Remove Power Supply Unit
    - Unplug all connected power cables.
    - Unscrew and carefully take out the power supply unit from the case.
10) Remove the Motherboard
    - Unscrew the motherboard from the case frame and gently lift it out.

Component Identification

Modern desktop computers contain numerous specialized components that serve specific functions within the overall system architecture. Understanding these components' purpose, specifications, and interconnections is essential for effective computer maintenance, troubleshooting, and upgrade procedures.

1) Motherboard

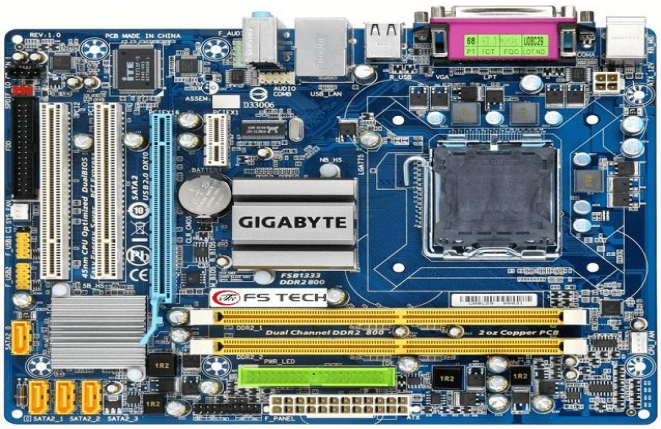


Figure 2 Motherboard

- It serves as the primary circuit board that connects and coordinates all system components. Modern motherboards feature complex multi-layer designs with integrated circuits, expansion slots, and connection interfaces that support current hardware standards.

- **CPU sockets** accommodate specific processor families and generations. These sockets feature hundreds of contact points that provide electrical connections between the processor and motherboard. Socket compatibility determines processor upgrade options and system performance potential.

Figure 3 CPU Socket

- **Memory slots** are physical connectors that hold RAM modules, allowing the system to access temporary storage for active programs and data. These slots vary in number and type, determining the maximum amount and speed of memory the system can support.

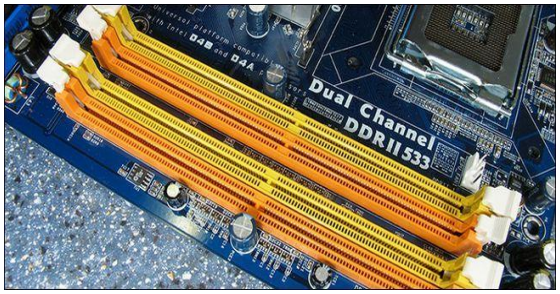


Figure 4 RAM slots

- **Expansion slots** include PCIe x16 slots for graphics cards, PCIe x1 slots for smaller expansion cards, like network cards, sound cards, etc. Older motherboards may still include PCI slots for legacy peripheral cards and AGP slots for dedicated graphics ports

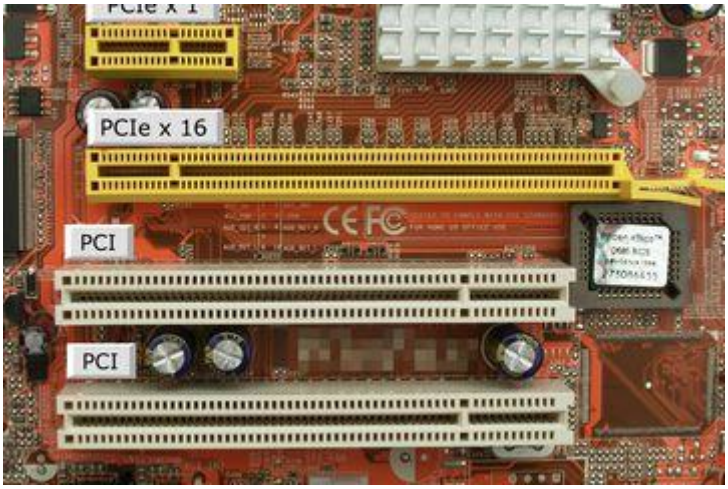


Figure 5 Expansion slots

- **Northbridge**: Handles high-speed communication between the CPU, RAM, and graphics components. It manages the front-side bus (FSB), memory controller, and PCIe connections, directly affecting system performance.

- **Southbridge**: Controls input/output functions and manages slower peripheral devices, including USB ports, SATA connections, audio, and legacy interfaces.

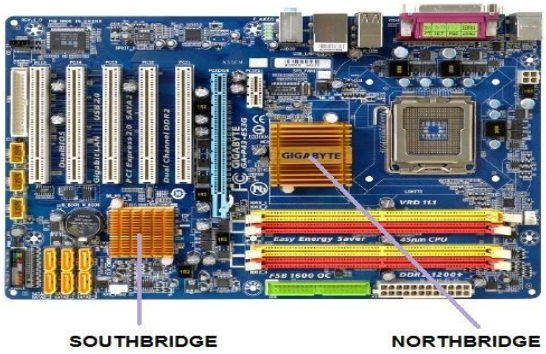


Figure 6 Northbridge/Southbridge

- The **BIOS** (Basic Input/Output System) or **UEFI** (Unified Extensible Firmware Interface) chip contains firmware that initializes hardware during startup and provides an interface for system configuration. UEFI is the modern replacement for BIOS, offering faster boot times, graphical interfaces, and enhanced security features.

- **CMOS Battery**: A small battery that maintains BIOS/UEFI settings and system time when the computer is powered off. Without this battery, system settings and system time would be lost each time power is disconnected.

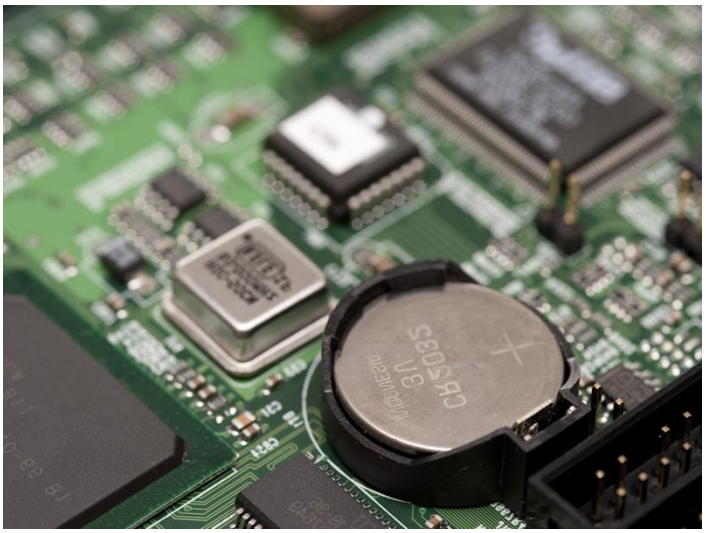


Figure 7 CMOS battery

- **Jumpers and DIP switches**: Small configuration devices used to set hardware options manually. Jumpers are metal bridges with plastic covers that connect pins to configure settings, while DIP switches are small toggle switches arranged in packages. These are less common on modern motherboards, as most settings are now configured through software.

- **I/O Port**: An I/O port, short for Input/Output port, is a physical interface on a computer or electronic device that allows data to be transferred between the device and external devices. It acts as a pathway for communication, enabling the exchange of data between the CPU and peripherals like keyboards, mice, printers, and storage devices.

- **USB Ports**: A USB (Universal Serial Bus) port is a standard connection point on computers and other electronic devices for connecting peripherals and transferring data or power.

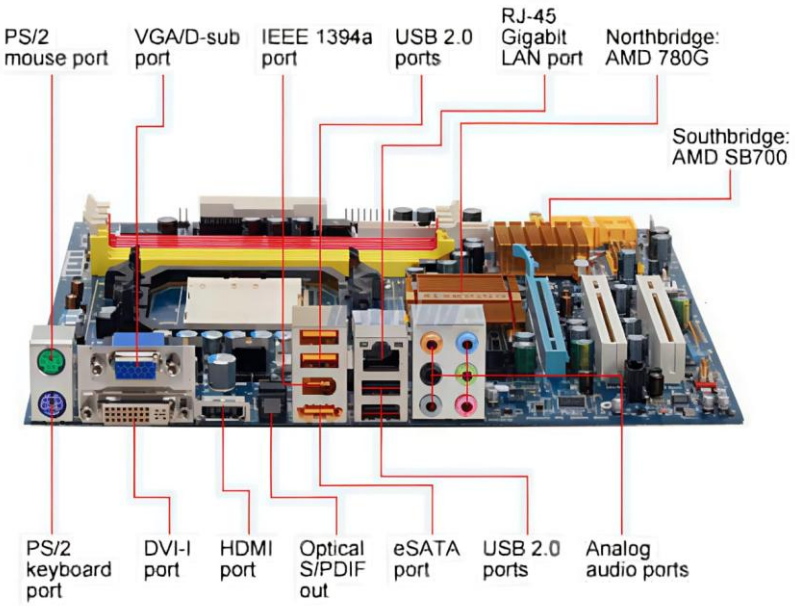


Figure 8 Different Ports on a Motherboard

- **VGA** (Video Graphics Array) is a standard analog display interface that connects video output devices to display devices such as monitors.

- **Ethernet Port**: An Ethernet port, also known as a LAN port or an RJ45 port, is a standard component found on most computer motherboards. It facilitates wired network connections.

- **PS/2 port**: The PS/2 port is a 6-pin mini-DIN connector used to connect keyboards and mice to a computer. It was introduced by IBM in 1987 as part of their Personal System/2 line of computers and became a standard interface throughout the 1990s and early 2000s.

- **DVI-I Port**: The DVI-I (Digital Visual Interface-Integrated) port is a DVI connector type that transmits both digital and analog video signals. DVI-I was designed to bridge the gap between older analog

and newer digital display technologies, offering a versatile video connectivity solution.

- **eSATA Port**: An eSATA (External Serial ATA) port is a connection interface on a computer or other device designed for connecting external storage devices, like Hard drives and SSDs. It's essentially an extension of the internal SATA interface, adapted for external use and offering high-speed data transfer capabilities.

- **HDMI Port**: HDMI (High-Definition Multimedia Interface) is a widely used digital audio/video interface that transmits uncompressed video and digital audio from a source device (like a set-top box or computer) to a compatible display device (like a TV or monitor) using a single cable. It supports resolutions from standard definition up to 10k, depending on the version.

- **Analog Audio Ports**: Analog audio ports are physical interfaces that transmit audio signals as continuously varying electrical voltages, directly representing the sound waves. Unlike digital ports, which convert audio into binary data, analog ports offer a simpler, more direct approach to audio connectivity.

- **IEEE 1394a port**: It is a standard for a high-speed serial bus, commonly known by brand names such as FireWire (Apple), and Lynx (Texas Instruments). It was a popular interface in the late 1990s and early 2000s, especially for connecting digital multimedia devices to computers. It was ideal for transferring large files due to its high data transfer speed.

- **Heat Sink:** Passive cooling devices made of thermally conductive materials (aluminium or copper) that dissipate heat from hot components like the CPU, chipset, and VRMs. They increase surface area to improve heat transfer to the surrounding air.

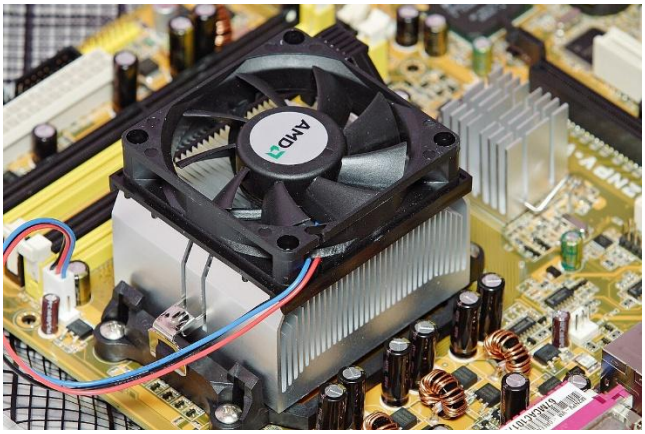


Figure 9 Heat Sink

- **Clock Generator**: Circuits that produce various clock frequencies needed by different system components.

2) RAM

Figure 10 RAM

- RAM (Random Access Memory) is a crucial component in any computing device, acting as the system's short-term, high-speed memory. It temporarily stores the data and instructions the processor (CPU) needs to access quickly to run applications and process information.

- Types of RAM: -

  a) **DRAM** (Dynamic RAM): Commonly used as main memory, it requires constant refreshing to retain data.

b) **SRAM** (Static RAM): Faster than DRAM and doesn't need refreshing, often used in processor caches.

c) **SDRAM** (Synchronous DRAM): Synchronizes with the system clock for more efficient data transfers.

d) **DDR SDRAM** (Double Data Rate SDRAM): An advanced form of SDRAM that transfers data twice per clock cycle, with generations like DDR2, DDR3, DDR4, and DDR5 offering improved performance.

e) **GDDRSDRAM** (Graphics Double Data Rate SDRAM): Designed for GPUs, optimized for high-speed data transfer for graphic tasks.

f) **NVDIMN** (Non-volatile Dual In-line Memory Module): Combines RAM speed with data retention when power is off.

- When programs or files are opened, data is loaded into RAM for quick access by the CPU. This enables smooth program execution and efficient multitasking. Data in RAM is erased upon system shutdown.

3) CPU (Central Processing Unit)

- The Central Processing Unit (CPU)—often referred to as the processor—is the primary component responsible for executing instructions and performing calculations within a computer. It acts as the "brain" of the system, interpreting program code, performing arithmetic and logical operations, and coordinating data flow between memory and peripherals.

Figure 11 CPU

- **Architecture and Cores**
  a) Modern CPUs use multicore architectures (e.g., dual-core, quad-core, hexa-core, octa-core) to handle multiple tasks or threads simultaneously.

  b) Each core contains its arithmetic logic unit (ALU) and control unit, enabling parallel instruction execution.

- **Clock Speed**
  a) Measure in gigahertz (GHz), clock speed indicates how many instruction cycles the CPU can perform per second.

  b) Higher clock speed generally translates to faster processing, though architectural efficiency and core count also affect overall performance.

- **Cache Memory**
  a) It is a small, high-speed memory component that stores copies of data from frequently used main memory (RAM) locations.

  b) Its primary purpose is to reduce the time and energy cost of accessing data from the main memory, thereby enhancing the overall speed and efficiency of the computer system.

- **Instruction Set**
  a) Defines the set of operations the CPU can execute (e.g., x84-64 for Intel and AMD desktop processors).

  b) Newer instruction extensions (SSE, AVX, etc.) accelerate specific tasks like multimedia processing and scientific computations.

- **Thermal Design Power (TDP)**
  a) Represents the average heat output under typical workloads, measured in watts (W).

  b) Indicates cooling requirements; higher TDP often necessitates more robust heatsinks or liquid-cooling solutions.

4) Storage Devices

- **Hard Disk Drive**
  A hard disk (or hard disk drive, HDD) is a non-volatile storage device that uses magnetically coated platters to record and retrieve digital data. Since the 1950s, HDDs have been foundational in computing, enabling persistent, large-scale storage for both consumer and enterprise applications**.**

Figure 12 Hard Disk Drive

- **Solid State Drive**

  Solid State Drives (SSDs) are a modern type of computer storage device that utilize integrated circuits to store data persistently. SSDs utilize NAND flash memory and a controller chip. Data is stored in semiconductor cells within the NAND flash memory, which are organized into blocks and pages. The controller manages data and employs techniques like wear leveling and garbage collection to help prolong the SSD's lifespan and maintain its performance.

Figure 13 Solid State Drive

Comparison

Feature	**HDD**	**SSD**
Technology	Magnetic platters	NAND flash memory
Capacity	Up to 22 TB per drive	Typically, ≤8 TB per module
Sequential Speed	80-250 MB/s	500-7000 MB/s
Latency	4-12 ms	< 0.1 ms
Power consumption	4-8 W active, 0.5-1.5 W idle	2-5 W active, <0.1 W idle
Use Case	Bulk Storage, backups, etc.	OS drive, databases, etc.

5) Graphics Card

A graphics card, also known as a video card, is a dedicated expansion board that renders images, video, and animations for a display device.

Unlike integrated graphics, which share system memory and reside on the CPU or motherboard, discrete graphics cards feature:

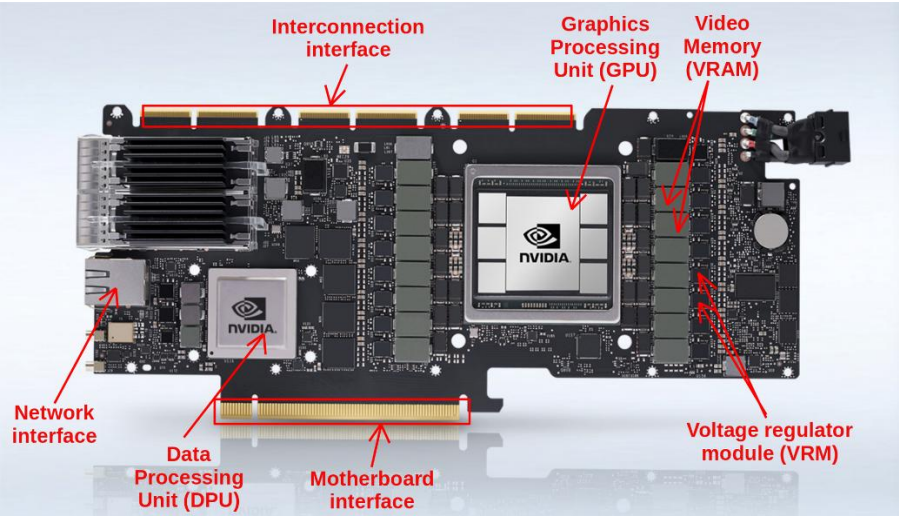


Figure 14 Graphics Card

- **GPU (Graphics Processing Unit):** A specialized processor designed primarily to accelerate computer graphics rendering and image processing. The most significant feature of GPUs is their parallel architecture, allowing them to perform many calculations at once.

- **VRAM (Video RAM):** It is a specialized type of RAM integrated directly onto a graphics card (GPU). It acts as a dedicated buffer between the GPU and the display, storing the images and graphics data that the GPU needs to render onto the computer screen.

- **Cooling System:** GPUs generate significant heat during operation, especially during demanding tasks like gaming or 3D rendering. The cooling system prevents the GPU from overheating and maintaining optimal performance. This typically consists of a large heatsink (metal

fins that absorb and dissipate heat) and one or more fans that blow air across the heatsink to further enhance cooling. Some high-end GPUs utilize liquid cooling systems.

- **Motherboard Interface:** This interface connects the graphics card to the computer's motherboard, allowing data and control signals to be exchanged between the GPU and the rest of the system. The dominant standard today is the PCI Express (PCIe) slot, which comes in various configurations (e.g., PCIe x16) offering different bandwidth levels.

- **Output Interfaces:** These ports on the graphics card allow you to connect a monitor or other display devices to the graphics card. Common output interfaces include HDMI, DisplayPort, DVI, and VGA.

- **Video BIOS (VBIOS):** Similar to the system BIOS (Basic Input Output System), the VBIOS contains a program that initializes and controls the graphics card during startup, storing information like clock frequencies, voltages, and fan speed control settings.

- **Power Connectors:** High-performance graphics cards often require more power than the PCIe slot alone can provide. Power connectors (e.g., 6-pin, 8-pin, or 12-pin) supply additional power directly from the power supply unit (PSU) to the graphics card.

- **Voltage Regulator Module (VRM):** The VRM ensures the GPU receives the correct and stable voltage, converting the PSU's 12V supply to the GPU's required operating voltage (typically around 1.1V).

6) Power Supply Unit (PSU)

A Power Supply Unit (PSU) is a crucial component in electronic systems, particularly computers, responsible for converting AC power from a wall outlet into the low-voltage DC power needed by computer components.

A stable power supply is essential for the functionality of electronic devices.

Types of Power Supplies: -

- Based on form factor:

a) **ATX (Advanced Technology Extended):** The standard form factor for most desktop computers, known for its widespread compatibility and availability. It typically measures 150 × 86 mm (width × height).
b) **SFX (Small Form Factor):** Designed for compact systems and mini-ITX cases. Smaller than ATX, with dimensions around 125mm wide, 63.5mm high, and 100mm deep.

c) **TFX (Thin Form Factor Extended):** Tailored for slim and low-profile desktop systems. It is more compact than ATX, with dimensions of 85 × 65 × 175 mm (width × height × length).

d) **EPS (Entry-Level Power Supply):** Primarily designed for high-performance workstations and servers. The EPS standard specifies higher wattage and stricter reliability compared to standard ATX PSUs. EPS power supplies have a 24-pin motherboard connector and an 8-pin or 8+8-pin 12V connector for demanding CPUs.

e) **Flex-ATX:** Designed for ultra-compact cases where space is severely limited.

- Based on modularity:

a) **Non-Modular PSU:** All cables are permanently attached to the unit, offering simplicity but potentially leading to cable clutter and limited customization options. These are generally more affordable.

b) **Semi-Modular PSU:** Includes fixed essential cables (like the 24-pin ATX connector) and detachable cables for others like PCIe and SATA power. This offers a balance between cost and flexibility.

c) **Fully Modular PSU:** All cables are detachable, providing the most flexibility for cable management, customization (e.g., custom sleeved cables), and upgrades. However, they are typically more expensive.

Reassembly of the Desktop computer

1) Install the Motherboard
   - Place the motherboard in the case and secure with screws
   - Attach the I/O shield if not already done.

2) Install the CPU
   - Insert the CPU into the socket using proper alignment
   - Secure with the socket lever.

3) Apply thermal paste and install heatsink
   - Apply a small amount of thermal paste on the CPU.
   - Attach the heatsink and secure it.
   - Plug the CPU fan cable into the motherboard header.

4) Install RAM
   - Insert RAM modules into their slots, pressing until the clips snap into place.

5) Install Storage devices
   - Mount SSDs and HDDs in their designated bays.
   - For M.2 drives, insert into the motherboard slot and secure with the screw.

6) Install Power Supply Unit (PSU)
   - Place the PSU in its location and secure it with screws.
   - Attach the required power cables.

7) Install Graphics Card and other expansion cards
   - Insert the graphics card in the appropriate PCIe slot and secure it.

8) Connect internal cables
   - Attach power cables to the motherboard, CPU, GPU, and drives.
   - Connect data cables such as SATA for storage devices.
   - Connect case cables (front panel, USB, audio) per case and motherboard instructions.

9) Cable management
   - Route and tie down cables to keep the interior tidy and maximize airflow.

10) Install System Fans
   - Secure fans to the case at intake and exhaust locations.
   - Connect fan cables to fan controller header.

11) Final Check
   - Ensure all components are securely installed.
   - Verify all cables are connected correctly.

12) Close the case
   - Reattach the side panels and secure with screws.

13) Connect External Peripherals
   - Attach monitor, keyboard, mouse, and other peripherals.

14) Power On and Test
   - Connect the power supply to an outlet and power on the PC.
   - Confirm system boots, and check for POST messages and component detection.

System Diagnosis Through BIOS/UEFI

After reassembling the desktop, a crucial step is to perform a system diagnosis using the BIOS (Basic Input/Output System) or its modern successor, UEFI (Unified Extensible Firmware Interface). This firmware is the first software that runs when you power on your computer, and it is responsible for initializing hardware and providing a platform for the operating system to load. Most BIOS/UEFI implementations include diagnostic tools that can help verify that the core components are functioning correctly before booting into the full operating system.

- Press Esc. key during the startup process to enter the BIOS/UEFI setup.
- Navigate to the Component Test section to diagnose hardware.

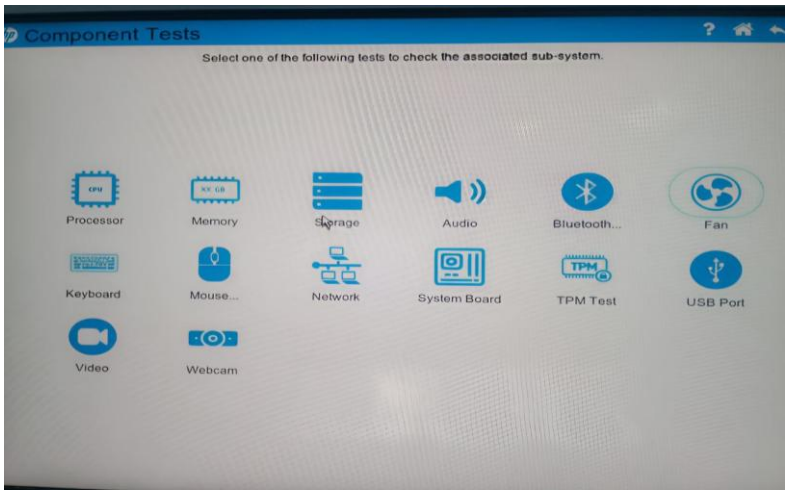
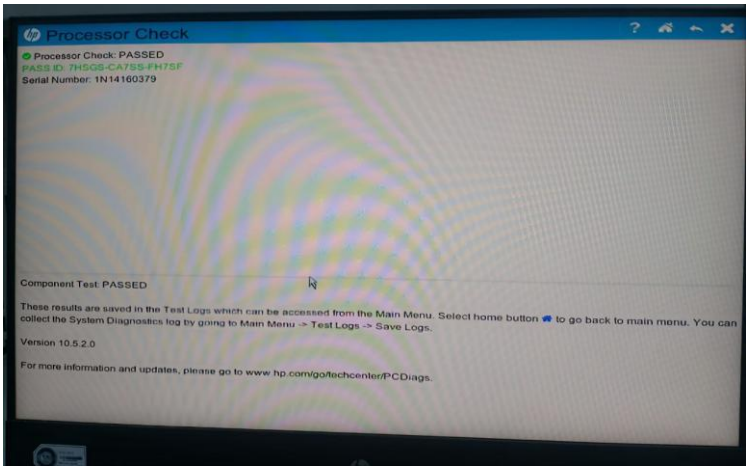


Figure 15 Diagnosis menu

1) CPU Test

- Confirms the processor is a genuine product.
- Verifies that the processor is operating at its correct clock speed.
- Pushes the CPU to its maximum processing load to check for stability and overheating issues. This can reveal problems with the CPU itself or the cooling system.



Fif. 16 CPU Test

2) Memory Test

- It performs various complex test patterns to find "stuck" bits, data corruption, and other failures in the RAM modules.
- Helps to isolate a faulty RAM stick. If errors are found, testing each module individually is recommended to identify the specific one that is failing.

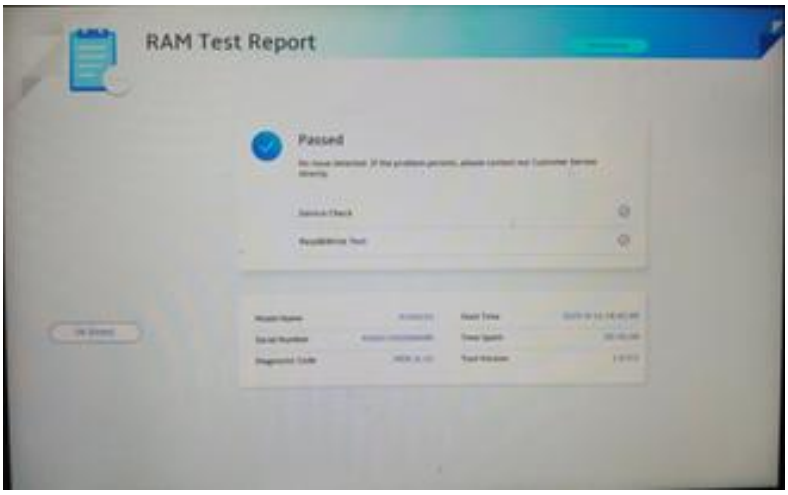


Figure 17 RAM Test

3) Storage Test

- Examine the drive's ability to read and write data.
- Queries the drive's built-in health status.

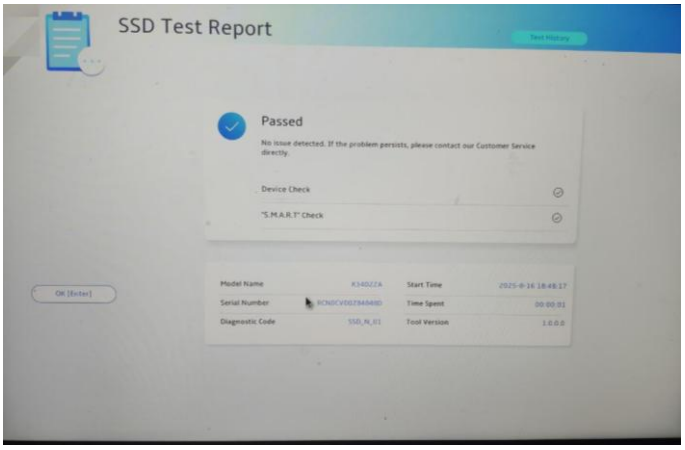


Figure 18 SSD Test

4) Fan Test

- Checks whether the fan is detected by the motherboard at all.
- Verifies the functionality and speed of the CPU and system fans.

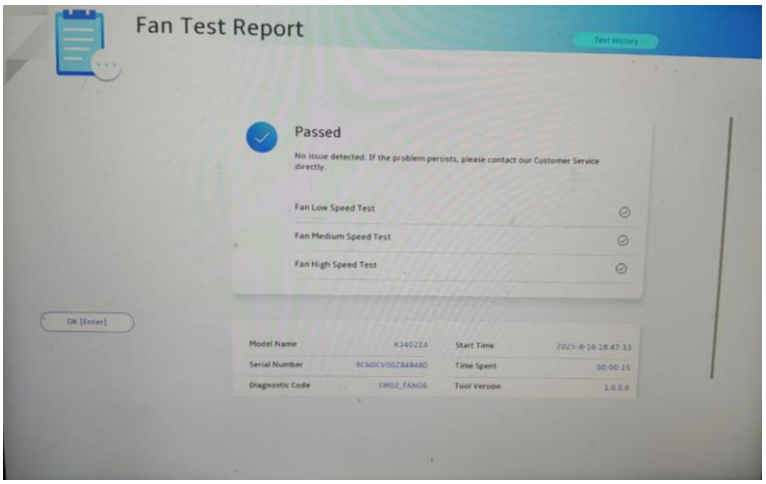


Figure 19 Fan Test

5) Keyboard Test

- Verifies the functionality of each key.
- Detects unresponsive or incorrect key mapping.

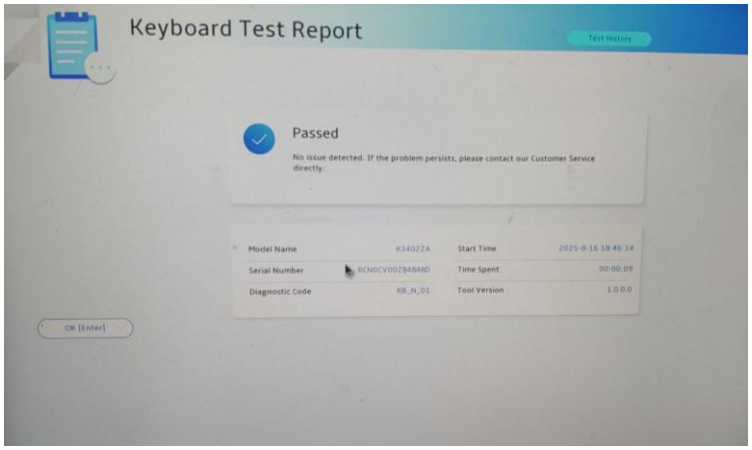


Figure 20 Keyboard Test

6) Audio Test

- Verifies the functionality of speakers and the microphone.
- Detects issues such as distortion, low volume, or no audio playback.

7) Mouse Test

- Ensures proper functionality of the mouse.
- Verifies pointer movements, button clicks, and scrolling.

8) Display Test

- Checks for dead pixels, colour accuracy, brightness, and constant.
- Ensures that images, text, and videos are displayed correctly without flickering or distortion.

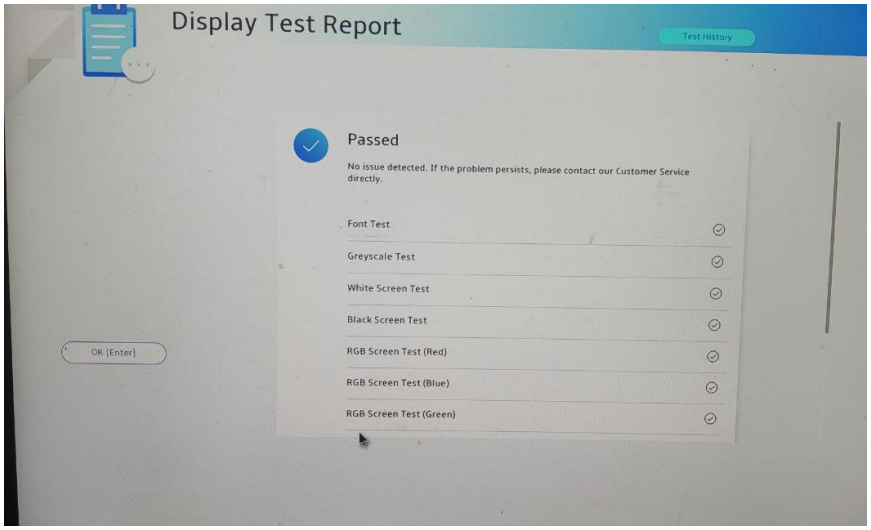


Figure 21 Display Test

9) USB Port Test

- Verifies the functionality of all USB ports on the system.
- Ensures proper data transfer between external devices and the computer.
- Detects issues such as loose connections, unrecognized devices, or slow transfer speeds.

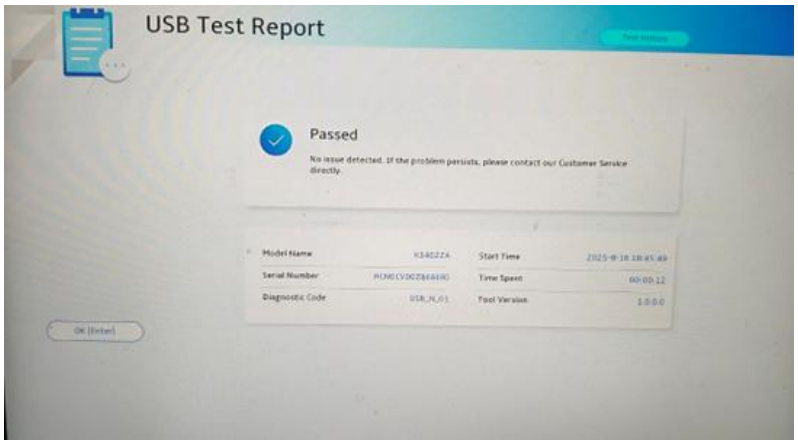


Figure 22 USB Test

10) Webcam Test

- Verifies the functionality of the built-in or external webcam.
- Detects issues such as no video signal, lagging, or distorted image output.

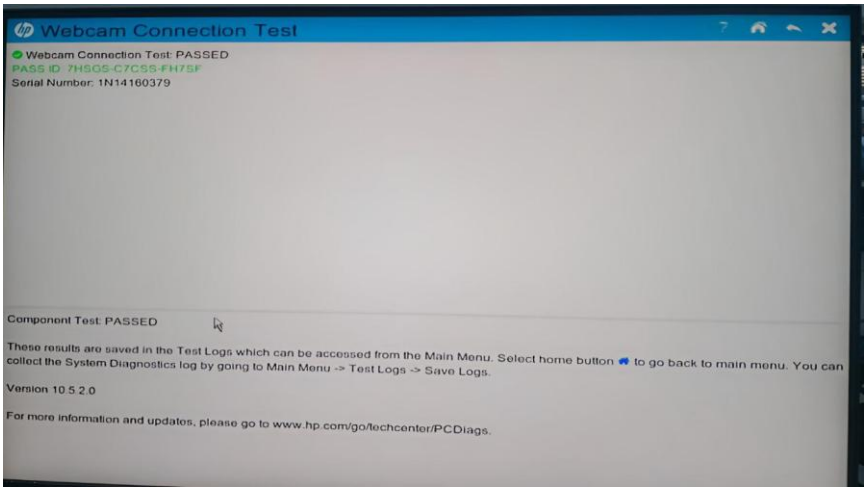


Figure 23 Webcam Test

The successful completion of this laboratory exercise demonstrates mastery of fundamental computer hardware concepts and practical assembly skills. The systematic approach to disassembly, identification, and reassembly, combined with comprehensive diagnostic testing, provides a solid foundation for understanding modern computer systems. This hands-on experience has significantly enhanced theoretical knowledge and developed practical skills essential for computer hardware maintenance and troubleshooting.

The exercise underscores the importance of methodical approaches, proper tool usage, and systematic verification procedures in computer hardware work, preparing students for advanced studies and professional applications in computer technology fields.

Job 2

Aim: Disassembly, Component Identification, and Reassembly of a laptop

Theory:

What is a laptop?

A laptop is a portable, all-in-one electronic device designed for mobile use. It integrates a screen, keyboard, touchpad, and rechargeable battery into a compact and lightweight form factor. Like a desktop computer, a laptop is composed of hardware and software that work in conjunction to process data, run applications, and connect to networks.

- Hardware: These are the physical components of the laptop, including the central processing unit (CPU), random-access memory (RAM), storage drives (SSD or HDD), display screen, keyboard, touchpad, battery, and motherboard that connects all the internal parts.
- Software: This includes the operating system (e.g., Windows, macOS, Linux) that manages the hardware and provides a user interface, as well as application software (e.g., web browsers, office suites, games) that allows users to perform specific tasks.

Tools Required to Disassemble a Laptop Computer

- Screwdrivers: A set of small Phillips-head and flathead screwdrivers is essential. Some laptops may also use Torx screws, requiring a corresponding screwdriver.
- Plastic Pry Tools: These are crucial for safely separating plastic case components without causing scratches or damage.
- Anti-static Wrist Strap: This protects sensitive electronic components from damage caused by electrostatic discharge.
- Container for Screws: A compartmentalized box or tray to keep screws organized is highly recommended, as laptops use various sizes of screws.

- Soft Brush: For cleaning dust from internal components like fans and heatsinks.

Disassembly of a Laptop computer

1. **Power Down and Disconnect:** Completely shut down the laptop and unplug the AC adapter.

2. **Remove the Battery:** If the laptop has an external, removable battery, take it out first. For internal batteries, you will need to open the case to disconnect it.

3. **Discharge Static Electricity:** Wear an anti-static wrist strap or touch a large metal object to ground yourself before handling internal components.

4. **Open the Case:** Place the laptop on a clean, flat surface. Remove the screws from the bottom panel of the laptop. Carefully use plastic pry tools to release the clips holding the bottom case in place and remove it.

5. **Disconnect the Internal Battery:** If the battery is internal, the first step after opening the case is to carefully disconnect its cable from the motherboard.

6. **Remove Storage Devices and RAM:** Locate and remove the storage drive (SSD or HDD) by disconnecting its cable and removing any securing screws. To remove RAM modules, gently press the clips on either side to release the module, then pull it out at an angle.

7. **Disconnect Other Components:** Systematically disconnect other components from the motherboard, such as the Wi-Fi card, cooling fan, keyboard, touchpad, and display cables. Pay close attention to the types of connectors and release mechanisms.

8. **Remove the Motherboard:** After all components are disconnected, carefully remove the screws securing the motherboard to the chassis. Gently lift the motherboard out of the case.

Component Identification

Modern laptops feature a highly integrated design where multiple components are housed in a compact space. Understanding these components is key to maintenance and upgrades.

- **Motherboard:** The main circuit board that connects all internal components, including the CPU, RAM, and storage. Laptop motherboards are designed to fit the specific form factor of the device.

Figure 1 Motherboard

- **CPU (Central Processing Unit):** The "brain" of the laptop that performs most of the processing. In many modern laptops, the CPU is soldered directly to the motherboard and is not user-replaceable.

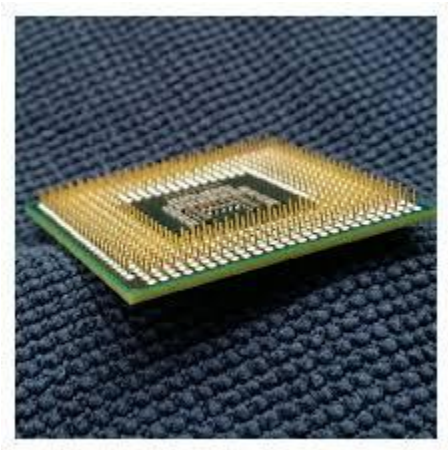


Figure 2 CPU

- **RAM (Random Access Memory):** Provides temporary storage for data and applications that are actively in use. Laptops typically use smaller SO-DIMM (Small Outline Dual In-line Memory Module) RAM sticks.

Figure 3 RAM

- **Storage Drive (SSD/HDD):** Stores the operating system, applications, and user files. Laptops may use either a traditional Hard Disk Drive (HDD) or a faster Solid-State Drive (SSD).

Figure 4 SSD

- **Cooling System:** Consists of a fan and a heatsink to dissipate heat generated by the CPU and sometimes the GPU. Keeping this system clean is crucial for performance.

Figure 5 Cooling System

- **Battery:** Provides power to the laptop when it is not plugged into an AC outlet.

Figure 6 Battery

- **Display Assembly:** Includes the LCD or LED screen, webcam, and Wi-Fi antennas.

Figure 7 Display

- **Keyboard and Touchpad:** The primary input devices for the laptop.

Figure 8 Keyboard & Touchpad

Reassembly of the Laptop computer

1. **Install the Motherboard:** Carefully place the motherboard back into the chassis and secure it with the correct screws.
2. **Reconnect Components:** Reconnect all the components to the motherboard in the reverse order of disassembly. This includes the display, keyboard, touchpad, cooling fan, and Wi-Fi card.
3. **Install RAM and Storage:** Re-insert the RAM modules into their slots and reinstall the storage drive, securing it and connecting its cable.
4. **Reconnect the Internal Battery:** Connect the internal battery cable back to the motherboard.
5. **Close the Case:** Reattach the bottom case, ensuring all clips snap into place. Secure it with the screws.
6. **Re-insert External Battery:** If applicable, re-insert the external battery.
7. **Final Check:** Ensure all connections are secure and that no screws are left over.

System Diagnosis Through BIOS/UEFI

After reassembly, it is important to perform a system diagnosis to ensure all components are functioning correctly. This can often be done through the laptop's BIOS (Basic Input/Output System) or UEFI (Unified Extensible Firmware Interface).

- **Accessing BIOS/UEFI:** To enter the BIOS/UEFI setup, you typically need to press a specific key (such as F2, F10, F12, or DEL) immediately after powering on the laptop.
- **Running Diagnostics:** Many modern laptops have built-in hardware diagnostic tools within the BIOS/UEFI.
- **Component Tests:**
    - **CPU Test:** Verifies the processor's functionality.

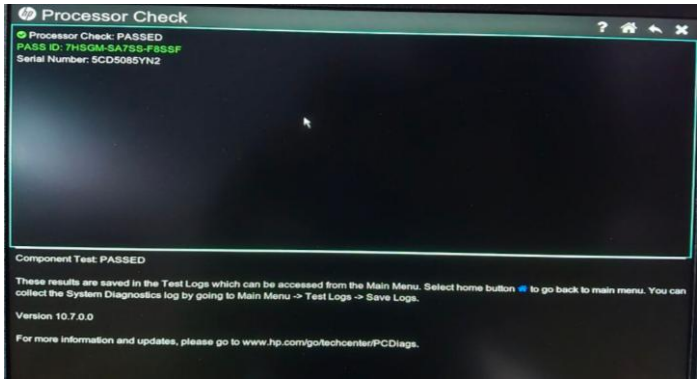


Figure 9 Processor Test

    - **Memory Test:** Checks the RAM for errors.

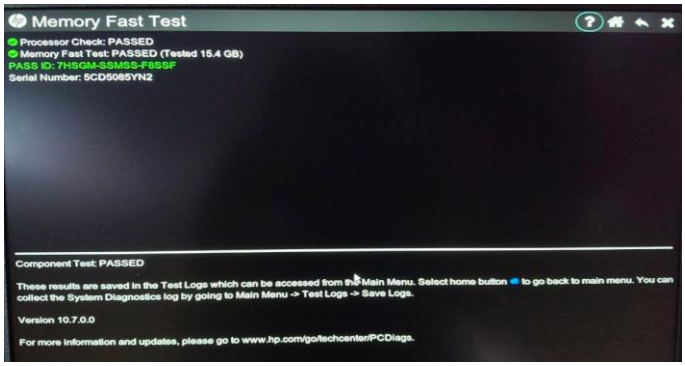


Figure 10 Memory Test

- **Fan Test:** Verifies the functionality and speed of the CPU and system fans.

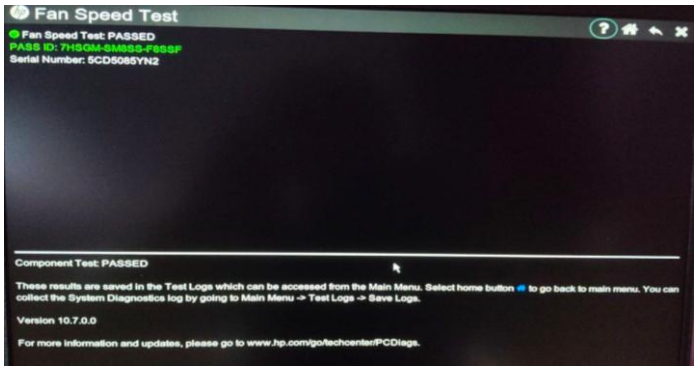


Figure 11 Fan Test

- **Keyboard Test:** Detects the unresponsive or incorrect key mapping.

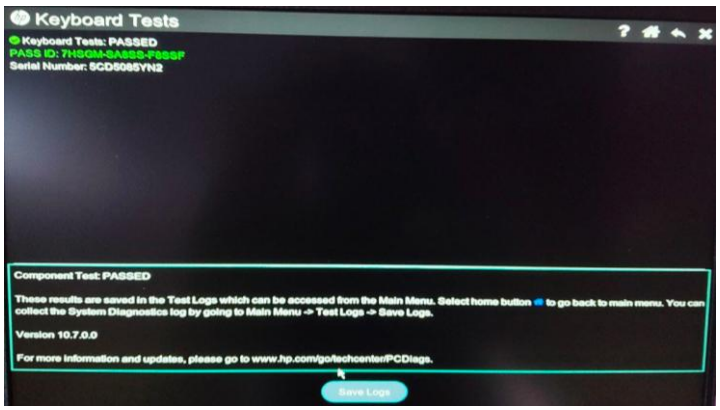


Figure 12 Keyboard Test

- **Touchpad Test:** Verifies pointer movements, button clicks, and scrolling.

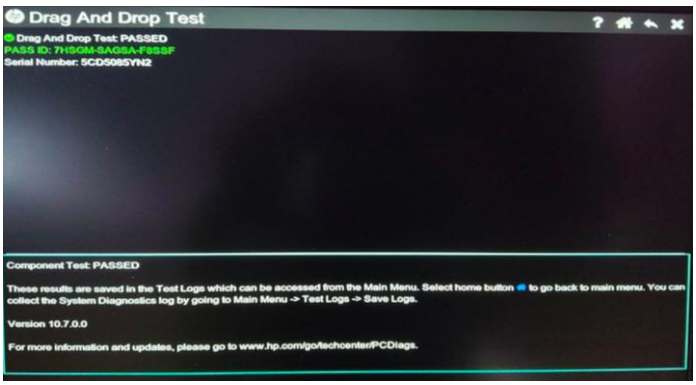


Figure 13 Touchpad Test

The successful completion of this laboratory exercise demonstrates a comprehensive understanding of laptop hardware, from its individual components to their intricate interplay. The methodical process of disassembly, identification, and reassembly, followed by diagnostic testing, provides a strong foundation for laptop maintenance, troubleshooting, and upgrades. This hands-on experience is invaluable for developing the practical skills necessary for a career in information technology and computer repair.

Job 3

Aim: To learn the essential features of Microsoft Word for creating, formatting, and finalizing professional documents.

Theory:

Introduction to Microsoft Office and Word

Microsoft Office is an integrated suite of business productivity applications developed by Microsoft. The cornerstone of this suite for document creation is Microsoft Word, a powerful and ubiquitous word processing application. It is used for creating, editing, and formatting a vast range of text-based documents, from simple letters and memos to complex academic reports, essays, technical manuals, and resumes. While it functions within a larger suite that includes spreadsheet and presentation software, this exercise will focus solely on the rich capabilities of MS Word as a standalone tool for effective written communication.

The Microsoft Word Interface

Microsoft Word is the most widely used word processor for creating professional-quality documents. Its interface is designed to be intuitive yet powerful, with a central "Ribbon" that organizes commands into logical tabs, making its extensive features accessible to users of all skill levels.

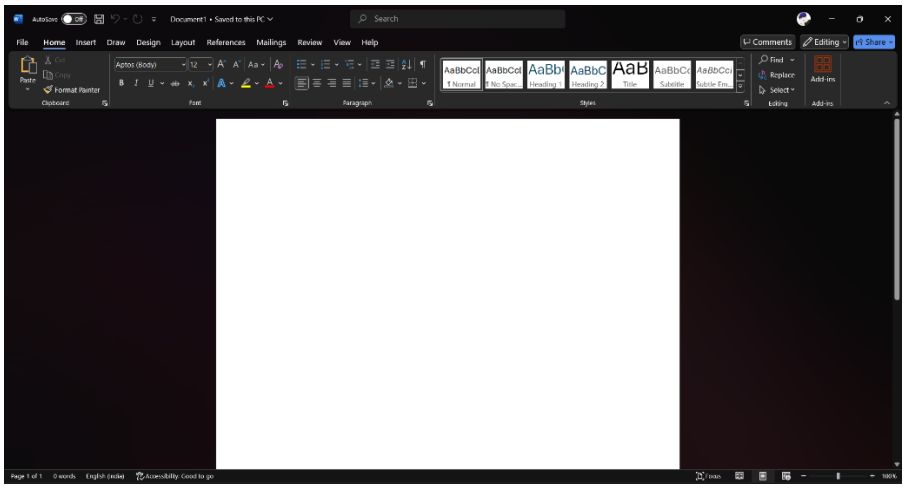


Fig. 1 MS Word Interface

- **The Ribbon:** Located at the top of the screen, the Ribbon is the primary command center. It is organized into task-oriented tabs (e.g., **Home** for basic formatting, **Insert** for adding objects, **Design** for

overall look, **Layout** for page setup, and **Review** for proofing). These tabs are further divided into logical groups of related commands, such as the "Font" and "Paragraph" groups on the Home tab. This design exposes users to the breadth of available functions in a structured way, preventing the interface from feeling cluttered or overwhelming.

- **The Document Area:** This is the main white space where you type, edit, and view your document's content. It is a "What You See Is What You Get" (WYSIWYG) editor, a crucial feature meaning the content on screen closely and accurately resembles how it will look when printed or converted to a PDF. This gives you precise control over the final appearance of your work.
- **The Status Bar:** Found at the bottom of the window, it provides useful, real-time information such as the current page number, total word count, proofing error status, and zoom controls. This allows for quick adjustments to the viewing layout and provides an at-a-glance summary of the document's state without interrupting your workflow.

Exploring Core Operations via the Ribbon Tabs

The Ribbon is the best way to understand the core operations of Word. Each tab is dedicated to a specific phase or type of document task.

1. The File Tab (Backstage View) This tab opens a full-page menu known as the Backstage view, where you manage the document as a whole.

- **Info:** Displays metadata about the document, such as file size and author. It also provides tools to protect the document with a password, inspect it for hidden data, and manage versions.
- **New:** Allows you to create a new blank document or choose from a wide variety of pre-designed templates like resumes, letters, and reports.
- **Open:** Used to open existing documents from your computer or cloud storage locations like OneDrive.
- **Save & Save As:** Save updates the current document with any changes. Save As is used to save the document with a new name, in a different location, or in a different file format (e.g., PDF, TXT, RTF).

- **Print:** Opens the print settings, where you can select a printer, set the number of copies, and preview how the document will look on paper.

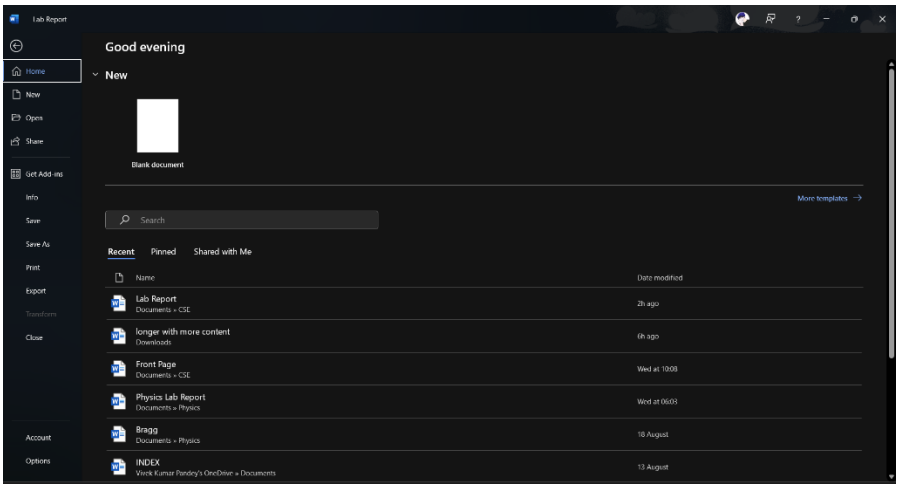


Fig. 2 File Tab

2. The Home Tab: The Essentials of Formatting This is the default and most-used tab, containing all the fundamental commands for shaping the appearance of your text and paragraphs.

- **Clipboard Group:** Contains the basic Cut, Copy, and Paste commands. It also includes the Format Painter, a tool that copies formatting from one selection of text to another.
- **Font Group:** This is your toolkit for character-level formatting. Beyond changing the font family, size, and color, you can apply styles like **bold**, *italic*, and <u>underline</u>. It also includes options for strikethrough, subscript (H_2O), and superscript (x^2), as well as text effects and highlighting.
- **Paragraph Group:** This group manages the structure and layout of paragraphs. You can set horizontal alignment (left, center, right, justify), control line and paragraph spacing, and create bulleted or numbered lists. It also allows you to apply borders and shading to make specific paragraphs stand out visually.
- **Styles Group:** One of Word's most powerful features, Styles are pre-defined sets of formatting that can be applied with one click. Using built-in styles like "Heading 1" and "Heading 2" not only ensures consistency but also structures your document logically.

- **Editing Group:** Includes tools to Find specific text, Replace text with an alternative, and Select all text or specific objects in your document.

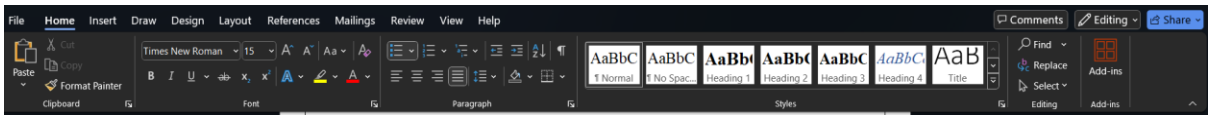


Fig. 3 Home Tab

3. The Insert Tab: Adding Content Beyond Text This tab is your gateway to enriching documents with non-textual elements.

- **Pages Group:** Allows you to insert a professional Cover Page, a Blank Page, or a Page Break to start new content on the next page.
- **Tables:** Allows you to insert a grid of rows and columns to organize data neatly. Once a table is created, contextual "Table Design" and "Layout" tabs appear, offering extensive control over the table's appearance and structure.
- **Illustrations:** This group lets you insert visual elements. You can add Pictures from your computer, use Shapes for simple diagrams, insert Icons, or create complex diagrams like flowcharts and organizational charts with SmartArt.
- **Header & Footer:** This feature lets you place repeating information at the top or bottom of each page, such as page numbers, document titles, or dates.
- **Text Group:** Allows you to insert a Text Box for content that needs to be positioned freely, add decorative text with WordArt, or create a large decorative first letter with Drop Cap.
- **Symbols Group:** Used to insert mathematical Equations or special characters and Symbols not found on the keyboard.

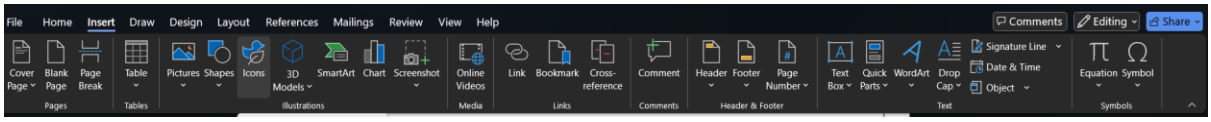


Fig. 4 Insert Tab

4. The Draw Tab: Inking and Annotation This tab is designed for freeform input, particularly on devices with a touch screen or stylus.

- **Drawing Tools:** You can select from a variety of pens and highlighters to draw, sketch, or handwrite annotations directly onto the document.
- **Ink to Shape/Math:** This intelligent feature can convert your rough hand-drawn shapes into perfect geometric figures or recognize handwritten mathematical formulas and convert them into editable, typed equations.

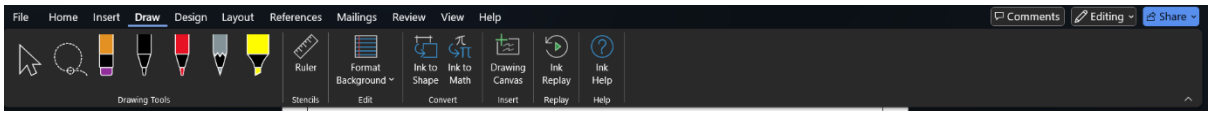


Fig. 5 Draw Tab

5. The Design Tab: Controlling the Overall Look This tab focuses on applying a consistent, professional look across the entire document.

- **Document Formatting:** This group includes Themes, which are professional combinations of colors, fonts, and effects. Applying a theme instantly changes the entire aesthetic of your document.
- **Page Background:** Here you can add a subtle text or image Watermark (e.g., "CONFIDENTIAL"), change the Page Color, or add Page Borders for a decorative touch.

Fig. 6 Design Tab

6. The Layout Tab: Page Structure and Arrangement This tab gives you high-level control over the physical layout of your document's pages.

- **Page Setup:** Define the page Margins, change the Orientation between Portrait (vertical) and Landscape (horizontal), and select the paper Size (e.g., A4, Letter). You can also format your text into newspaper-style Columns.
- **Breaks:** These tools control where text flows. A Page Break ends the current page and starts a new one. Section Breaks are more powerful; they divide your document into sections that can be formatted independently.
- **Arrange Group:** When working with images or shapes, this group lets you control their Position on the page, how text wraps around

them (Wrap Text), and their layering order (Bring Forward/Send Backward).

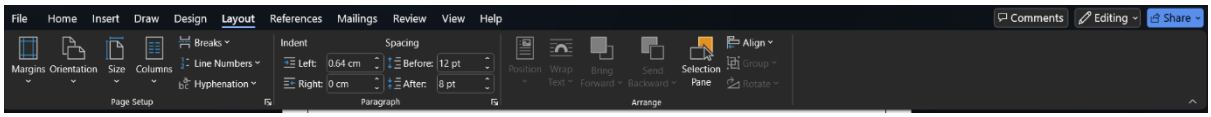


Fig. 7 Layout Tab

7. The References Tab: For Academic and Formal Writing This tab is indispensable for creating structured, long-form documents like research papers and reports.

- **Table of Contents:** Automatically generates a navigable table of contents based on the Heading styles you have applied.
- **Footnotes Group:** Lets you insert Footnotes (notes at the bottom of the page) or Endnotes (notes at the end of the document) for citations or extra information.
- **Citations & Bibliography:** A powerful tool to manage your sources, insert in-text citations in various academic styles (like APA or MLA), and automatically generate a formatted bibliography page.
- **Captions Group:** Insert numbered captions for figures, tables, and equations. This also allows you to automatically generate a list of figures or tables.

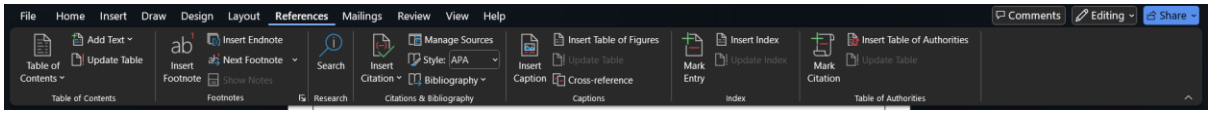


Fig. 8 Reference Tab

8. The Mailings Tab: For Mass Communication This tab is built around a single, powerful feature for creating personalized mass mailings.

- **Mail Merge:** This feature automates the creation of personalized letters, envelopes, or labels. It works by combining a main document (the template) with a data source (like an Excel spreadsheet or contact list), inserting unique recipient information into each copy.

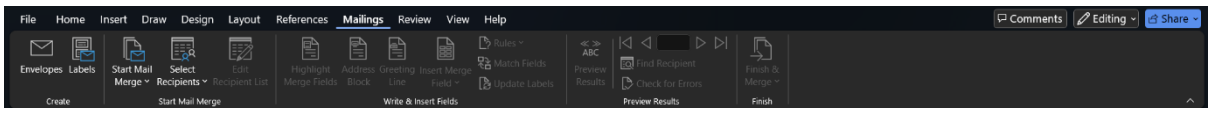


Fig. 9 Mailings Tab

9. The Review Tab: Proofing and Collaboration This tab provides tools for polishing your document and working with others.

- **Proofing:** Includes the essential Spelling & Grammar checker to catch errors. It also contains a Thesaurus for finding alternative words and a Word Count tool.
- **Comments Group:** Allows you to add, navigate, and delete comments in the margins, which is useful for providing feedback.
- **Tracking Group:** The Track Changes feature records every change made to the document. Each change can then be reviewed and either accepted or rejected, making it an essential tool for collaborative editing.

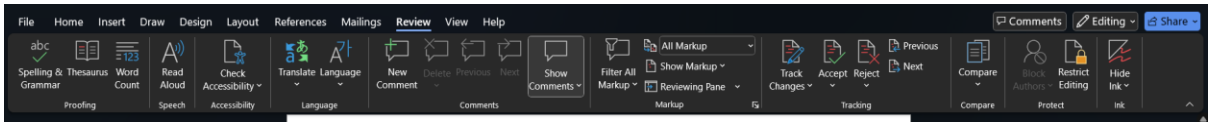


Fig. 10 Review Tab

10. The View Tab: Changing Your Perspective This tab lets you change how you look at your document without affecting its content or layout.

- **Views:** You can switch between different viewing modes, such as Read Mode, Print Layout, Web Layout, Outline, and Draft.
- **Show Group:** You can choose to show or hide helpful on-screen tools like the Ruler for setting tabs and indents, Gridlines for aligning objects, and the Navigation Pane for jumping between headings.
- **Window Group:** Contains tools to view multiple documents side-by-side or split the current document into two scrollable panes.

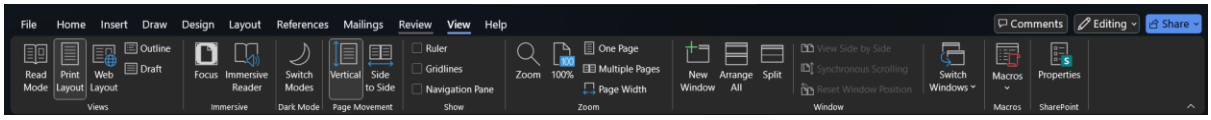


Fig. 11 View Tab

Essential Keyboard Shortcuts

Keyboard shortcuts can significantly increase efficiency when working in Word. Here are some of the most common and useful ones:

- **Basic Editing**
  - Ctrl + C: Copy selected text
  - Ctrl + X: Cut selected text
  - Ctrl + V: Paste text
  - Ctrl + Z: Undo the last action
  - Ctrl + Y: Redo the last action
  - Ctrl + F: Open the Find dialog box
  - Ctrl + S: Save the current document
- **Text Formatting**
  - Ctrl + B: Apply or remove bold formatting
  - Ctrl + I: Apply or remove italic formatting
  - Ctrl + U: Apply or remove underline formatting
  - Ctrl +]: Increase font size by one point
  - Ctrl + [: Decrease font size by one point
- **Paragraph Formatting**
  - Ctrl + L: Left-align paragraph
  - Ctrl + E: Center-align paragraph
  - Ctrl + R: Right-align paragraph
  - Ctrl + J: Justify paragraph alignment

This exercise confirms that Microsoft Word is a comprehensive tool for professional document production. Mastering its core functionalities, from formatting and layout to integrating tables and references, is essential for creating polished, structured documents. The skills developed here provide a robust foundation for producing clear and effective work for any academic or professional challenge.

Job 4

Aim: To create a flowchart in Microsoft Word illustrating the process of federated learning.

Theory:

Flowcharts are diagrams that represent a workflow or process. They use a set of standard symbols to depict various steps, decisions, and actions, connected by arrows that show the flow and sequence. In Microsoft Word, a **Drawing Canvas** (Insert > Shapes > New Drawing Canvas) is a useful tool for creating flowcharts, as it groups all the elements together, making them easier to move and resize as a single object.

The flowchart below illustrates the concept of Federated Learning. This is a machine learning approach that trains an algorithm across multiple decentralized devices or servers holding local data samples, without exchanging the data itself.

Steps for Creation:

1. **Insert a Drawing Canvas:** Go to the "Insert" tab, click on "Shapes," and select "New Drawing Canvas" at the bottom of the menu. This will create a bordered area to build your flowchart within.

2. **Create the Central Server:** From the "Shapes" menu, select the "Cloud" shape under "Basic Shapes." Draw it at the top-center of the canvas and add the text "Central Server."

3. **Add Client Areas:** Below the server, on the left and right, draw two large "Oval" shapes. Add text boxes inside them labeled "Area A" and "Area B."

4. **Represent Data Samples:** Inside each oval, insert smaller shapes like rectangles and circles to represent the "Different samples" as shown in the original image. Use the brace shape from "Basic Shapes" to group them and add a text box with the label "Different samples" underneath each area.

5. **Illustrate Aggregation:** Between the central server and the two areas, draw a few small, filled circles to represent the "Aggregation" process.

6. **Draw Connecting Arrows:**
    - Use the "Line Arrow" shape to draw lines from "Area A" and "Area B" pointing upwards towards the "Aggregation" point.
    - Draw arrows from the "Aggregation" point to the "Central Server." Add a text box next to these lines and label it "Local model."
    - Draw an arrow from the "Aggregation" point downwards and label it "Global model."
    - From the "Global model" line, draw arrows pointing to both "Area A" and "Area B."
7. **Add the "No Exchange" Symbol:** Between "Area A" and "Area B," insert an "Oval" shape. Select the "Line" shape and draw it diagonally through the oval. Select both the oval and the line, right-click, and choose "Group" to combine them.
8. **Finalize Formatting:** Adjust the colors, line styles, and text fonts using the "Shape Format" tab to complete the flowchart.